

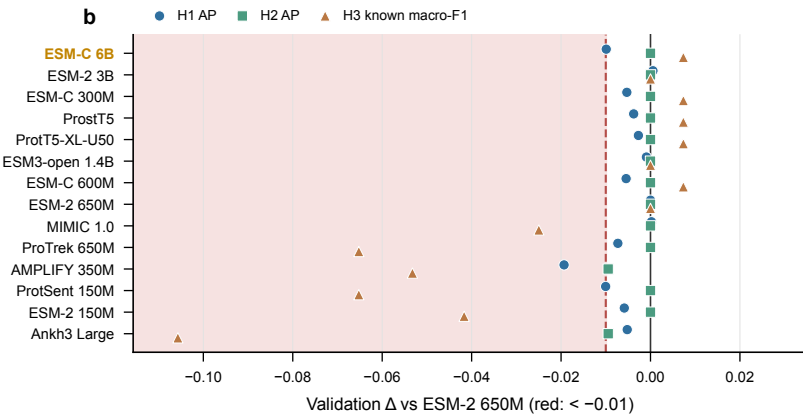
Development-only selection of protein representations

Frozen Train-only 5-fold global-component CV (mean $\pm$ fold SE)

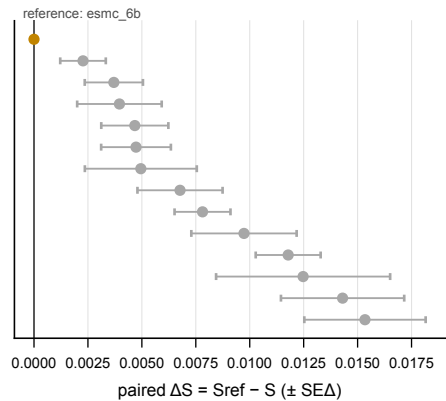
data-curation V3 (560 VMA-DJR) $\rightarrow$ project V0 · raw-score-stable-calibration-v1 · Train CV + Validation only · no Test metrics

	S	H1 AP	H2 AP	H3 known macro-F1
ESM-C 6B	0.997	0.998	1.000	0.981
ESM-2 3B	± 0.001	± 0.000	± 0.000	± 0.010
ESM-C 300M	0.995	0.999	1.000	0.953
ProstT5	± 0.001	± 0.000	± 0.000	± 0.007
ProtT5-XL-U50	0.993	0.995	1.000	0.962
ESM3-open 1.4B	± 0.002	± 0.001	± 0.000	± 0.017
ESM-C 600M	0.993	0.996	1.000	0.957
ESM-2 650M	± 0.002	± 0.003	± 0.000	± 0.009
MIMIC 1.0	0.992	0.997	1.000	0.930
ProTrek 650M	± 0.002	± 0.001	± 0.000	± 0.013
AMPLIFY 350M	0.992	0.995	1.000	0.952
ProtSent 150M	± 0.001	± 0.002	± 0.000	± 0.008
ESM-2 150M	0.989	0.993	1.000	0.963
Ankh3 Large	± 0.003	± 0.003	± 0.000	± 0.016
	0.990	0.998	1.000	0.916
	± 0.002	± 0.001	± 0.000	± 0.025
	0.989	0.984	1.000	0.930
	± 0.002	± 0.001	± 0.000	± 0.016
	0.987	0.995	1.000	0.907
	± 0.003	± 0.002	± 0.000	± 0.028
	0.985	0.985	0.999	0.866
	± 0.002	± 0.001	± 0.001	± 0.022
	0.985	0.988	1.000	0.919
	± 0.004	± 0.004	± 0.000	± 0.024
	0.983	0.991	1.000	0.879
	± 0.003	± 0.004	± 0.000	± 0.018
	0.982	0.991	1.000	0.872
	± 0.002	± 0.002	± 0.000	± 0.014

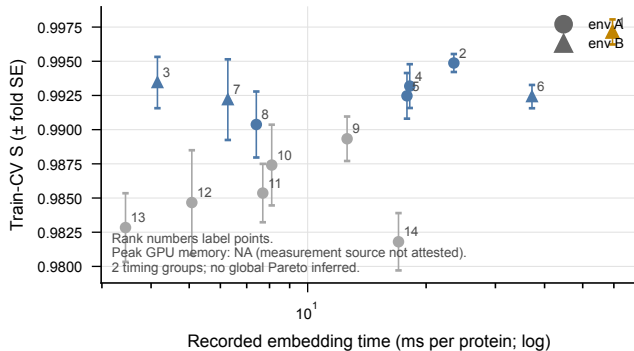
Validation gates



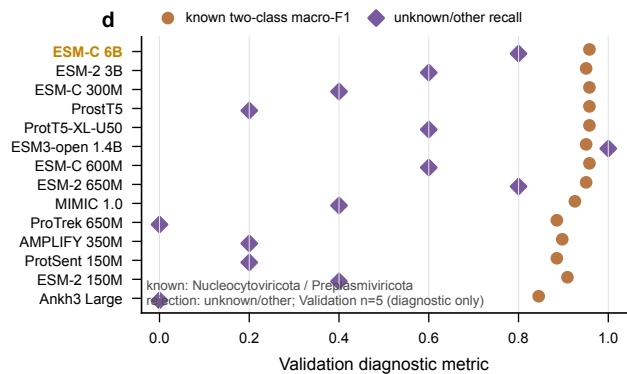
Paired one-SE



S-compute evidence (descriptive)



H3 output and rejection scope



Frozen decision audit — selected ESM-C 6B (baseline = ESM-2 650M)

Model	1-SE	mean $\Delta S \pm SE_{\Delta}$	Val gate	License	Pretraining-overlap risk	Decision
ESM-C 6B	yes	$\pm 0.000 \pm 0.000$	PASS	MIT	not flagged	selected; tie-break rank 1
ESM-2 3B	no	$\pm 0.002 \pm 0.001$	PASS	MIT	not flagged	outside paired one-SE set
ESM-C 300M	no	$\pm 0.004 \pm 0.001$	PASS	MIT	not flagged	outside paired one-SE set
ProstT5	no	$\pm 0.004 \pm 0.002$	PASS	MIT	high	outside paired one-SE set
ProtT5-XL-U50	no	$\pm 0.005 \pm 0.002$	PASS	Apache-2.0	not flagged	outside paired one-SE set
ESM3-open 1.4B	no	$\pm 0.005 \pm 0.002$	PASS	MIT	low (viruses excluded)	outside paired one-SE set
ESM-C 600M	no	$\pm 0.005 \pm 0.003$	PASS	MIT	not flagged	outside paired one-SE set
ESM-2 650M	no	$\pm 0.007 \pm 0.002$	PASS	MIT	not flagged	outside paired one-SE set
MIMIC 1.0	no	$\pm 0.008 \pm 0.001$	FAIL H3	MIT	high	excluded by Validation H3
ProTrek 650M	no	$\pm 0.010 \pm 0.002$	FAIL H3	MIT	high	excluded by Validation H3
AMPLIFY 350M	no	$\pm 0.012 \pm 0.002$	FAIL H1,H3	MIT	not flagged	excluded by Validation H1,H3
ProtSent 150M	no	$\pm 0.012 \pm 0.004$	FAIL H1,H3	MIT	high	excluded by Validation H1,H3
ESM-2 150M	no	$\pm 0.014 \pm 0.003$	FAIL H3	MIT	not flagged	excluded by Validation H3
Ankh3 Large	no	$\pm 0.015 \pm 0.003$	FAIL H3	CC-BY-NC-SA-4.0	not flagged	excluded by Validation H3